



FAPS

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Friedrich-Alexander-Universität
Technische Fakultät

Mechanical rotor design and production engineering calculations for stator windings

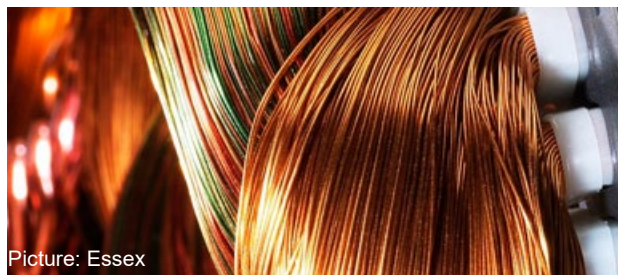
Vorlesung Produktion elektrischer Motoren und Maschinen
Lecture Production of electric motors and machines

A large number of design and manufacturing aspects must be taken into account on both the stator and rotor sides.

In order to understand the issues of electrical mechanical engineering, selected aspects are explained using examples

After attending the lecture unit "Production engineering calculations for stator windings and mechanical rotor design" you should:

- **Material properties and tolerance influences of the semi-finished wire product,**
- **Presentation forms of winding variants and**
- **better understand the qualification of rotor-side adhesive bonding technologies and be able to assess them using physical parameters.**



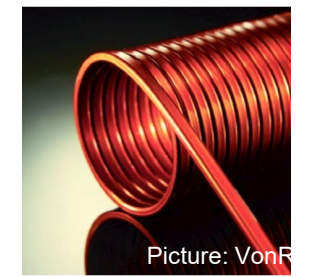
Picture: Essex



Picture: VonRoll



Picture: VonRoll



Picture: VonRoll

Part 1 (Rotor) of the specialisation builds on the understanding of lecture unit 4.

Relevant topics for preparation in self-study

Rotor assembly

Application example:

- Design of an adhesive joint for an IPM rotor

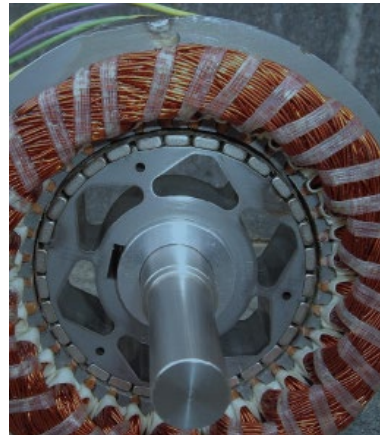
Methods:

- Strength of materials
- Dynamics

Technical functions:

- Fuses for high-speed concepts
- Qualification of connection technologies

Lecture unit
4



Hard magnetic rotor of a permanent magnet synchronous machine

Image: Swiss Federal Office of Energy SFOE (Switzerland)

Stator winding

Relevant topics:

- Filling factors
- Semi-finished wire products
- Winding specification
- Winding scheme
- Winding process
- Wire tension control

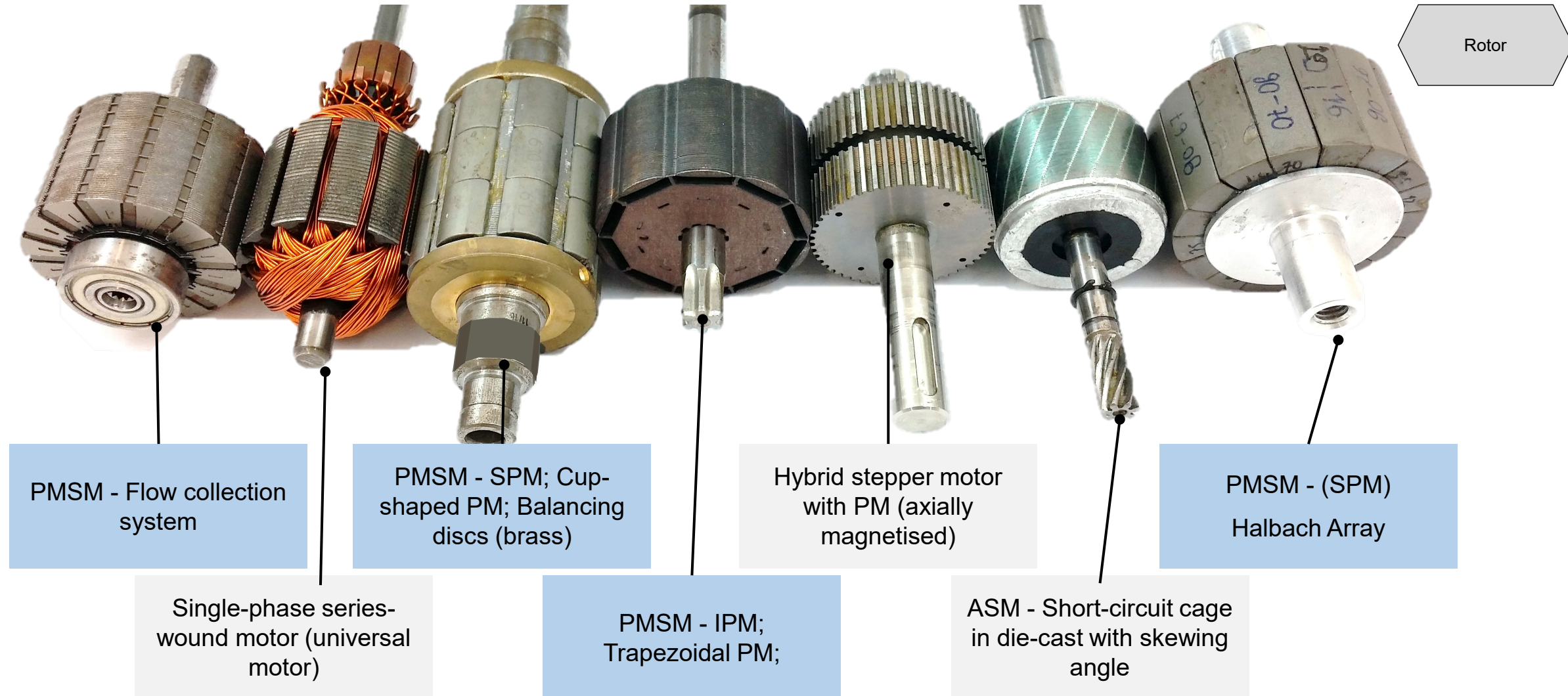
Lecture unit
5+6



Stator of a PSM with pull-in winding in automotive engineering

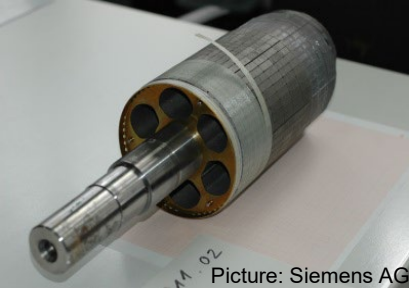
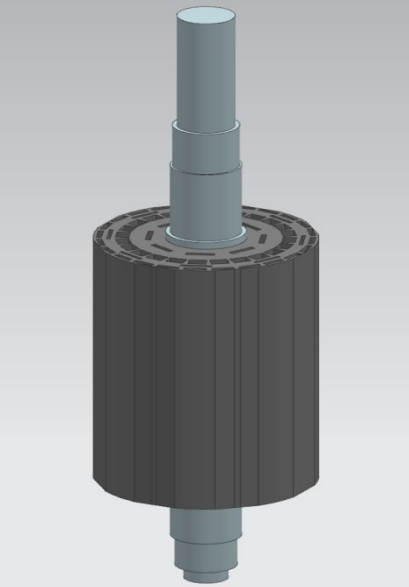
Picture: BMW

Permanent magnets are used within the conventional design
are used for various motor types.



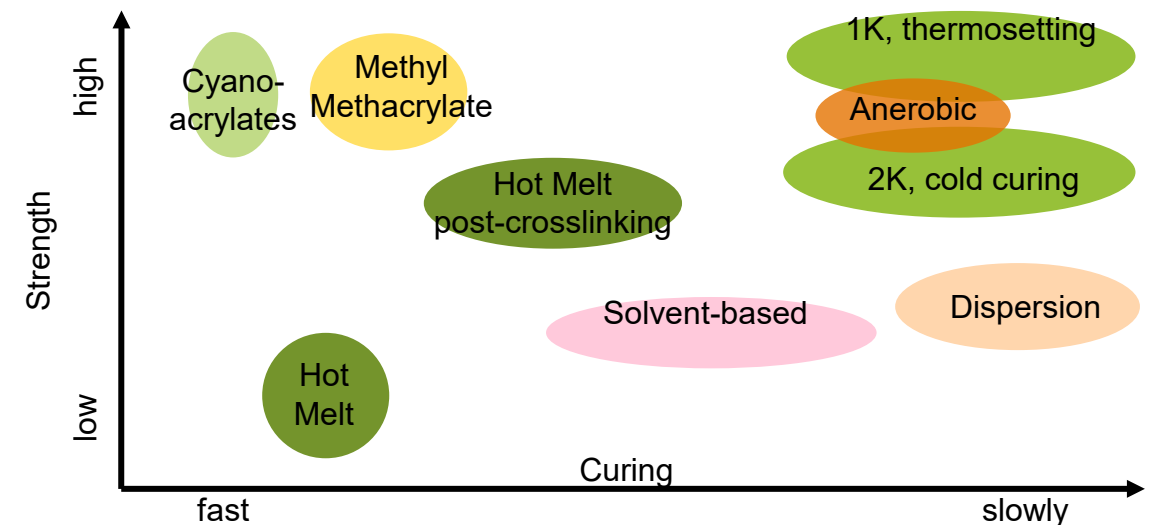
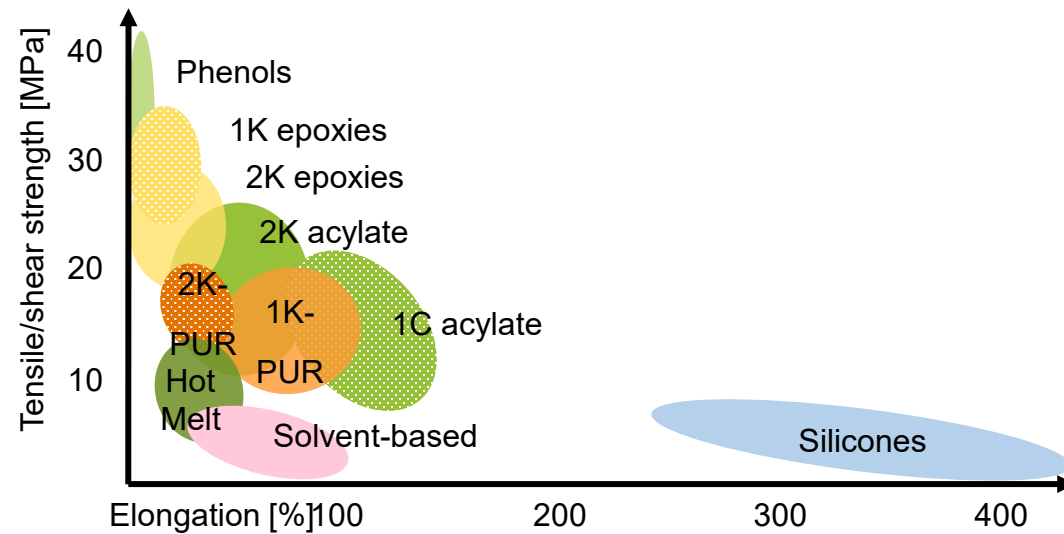
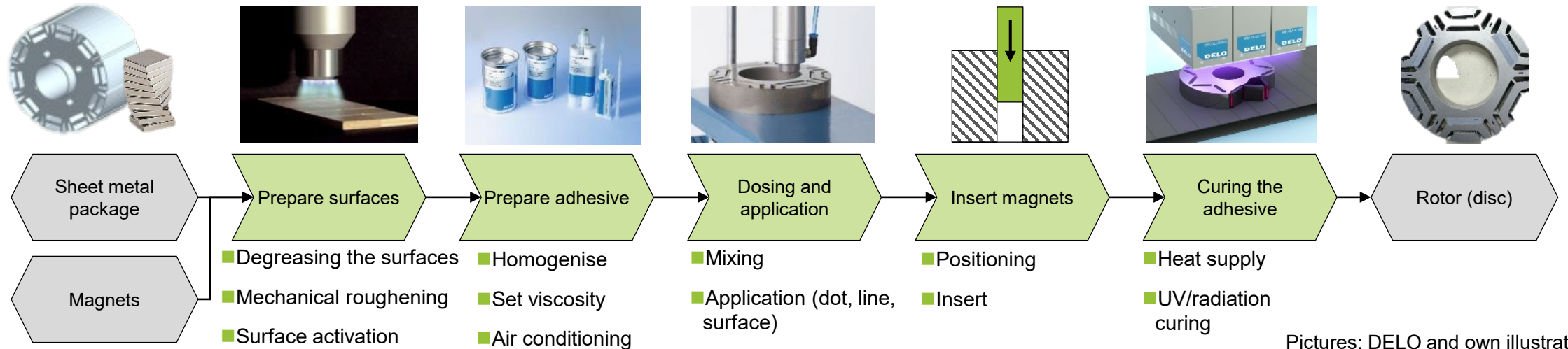
The rotor converts a magnetic torque into a mechanical torque and transmits this to the shaft. A PM rotor consists of the following components.

Slide from VE1

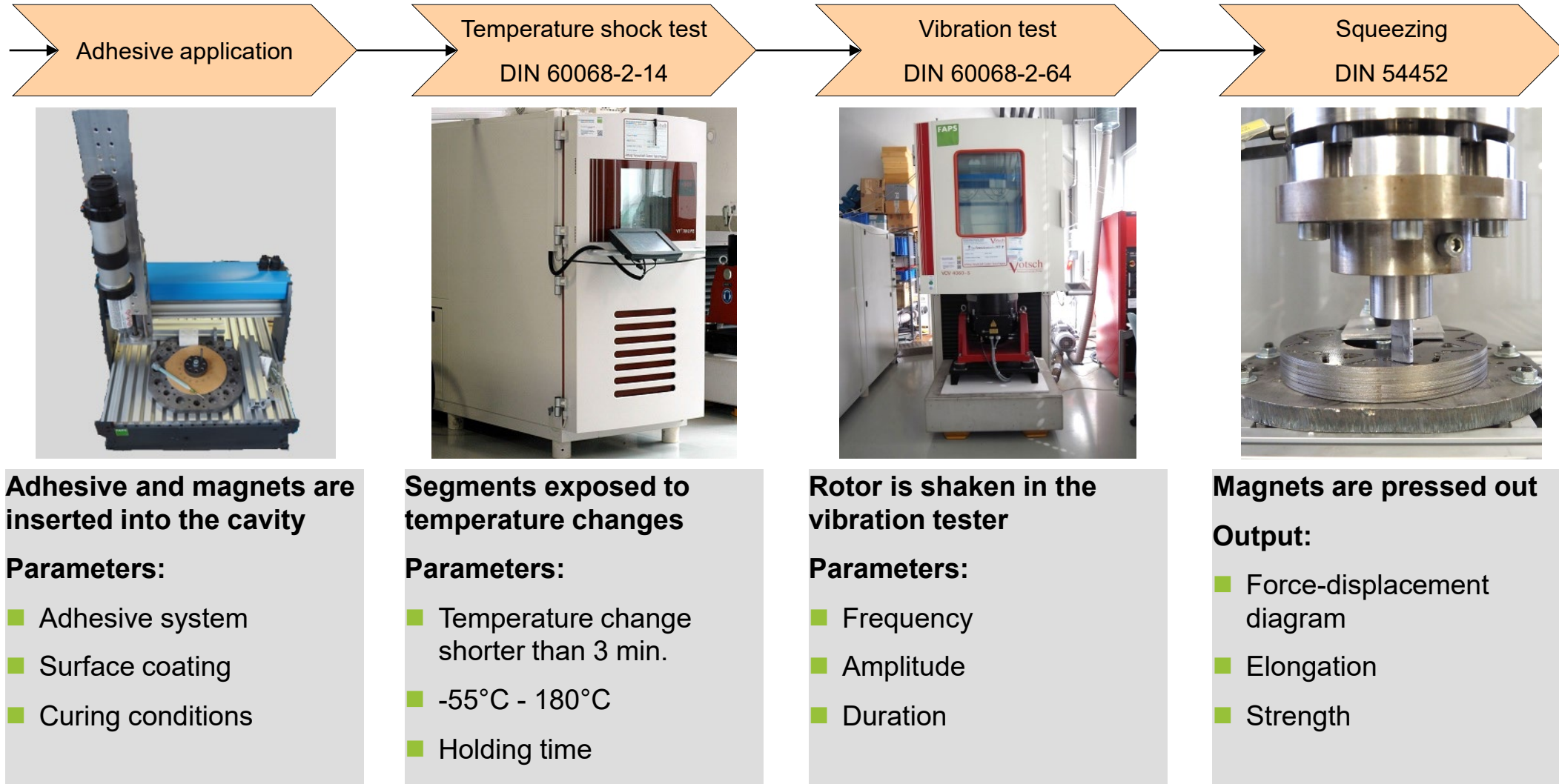
Assembly group	Component	Physical function
Rotor  	Rotor shaft	- Mechanical transmission of the torque - Ensuring the mechanical strength of the sheet metal package
	Rotor lamination stack	- Guidance of the magnetic flux with minimal losses - For IPM motors: Protection of the magnets against mechanical damage
	Permanent magnets	Formation of magnetic poles to generate the mechanical torque
	Bandaging	- Mechanical securing of the magnets - Protection of the magnets against mechanical damage
	End plates	- Correction of imbalances - Stabilisation of the sheet stack by applying an axial pressing force
	Insulating resin/ Adhesive	- Protection of the magnets against external influences (corrosion) - Positioning of the magnets in a fixed position (especially high speeds)

Rotor

The bonding of magnets has established itself as a standard process and is therefore considered as an example.



For the validation of adhesive bonds various methods are possible.



In the cavity, the bonding surfaces are subjected to tension or pressure.

The shear force F_{shear} and compressive/tensile force $F_{\text{tens,comp}}$ occur, whereby the centrifugal force F_c acts on the magnets at an angle of 55° .

Centrifugal force on shell element

$$F_c =$$

$$m_{\text{magnet}} =$$



The force components $F_{t,c}$ and F act on the solenoid.

$$F_{\text{tens/comp}} =$$

$$F_{\text{shear}} =$$

Magnet and pocket data:

- Dimensions Magnet: $b = 14,23 \text{ mm}$; $h = 4 \text{ mm}$; $l = 21,3 \text{ mm}$
- Density Magnet $\rho_{\text{equiv}} = 7600 * 10^{-9} \frac{\text{kg}}{\text{mm}^3}$
- Distance pivot point: $r = 65 \text{ mm}$
- Speed $n = 15000 \text{ rpm}$

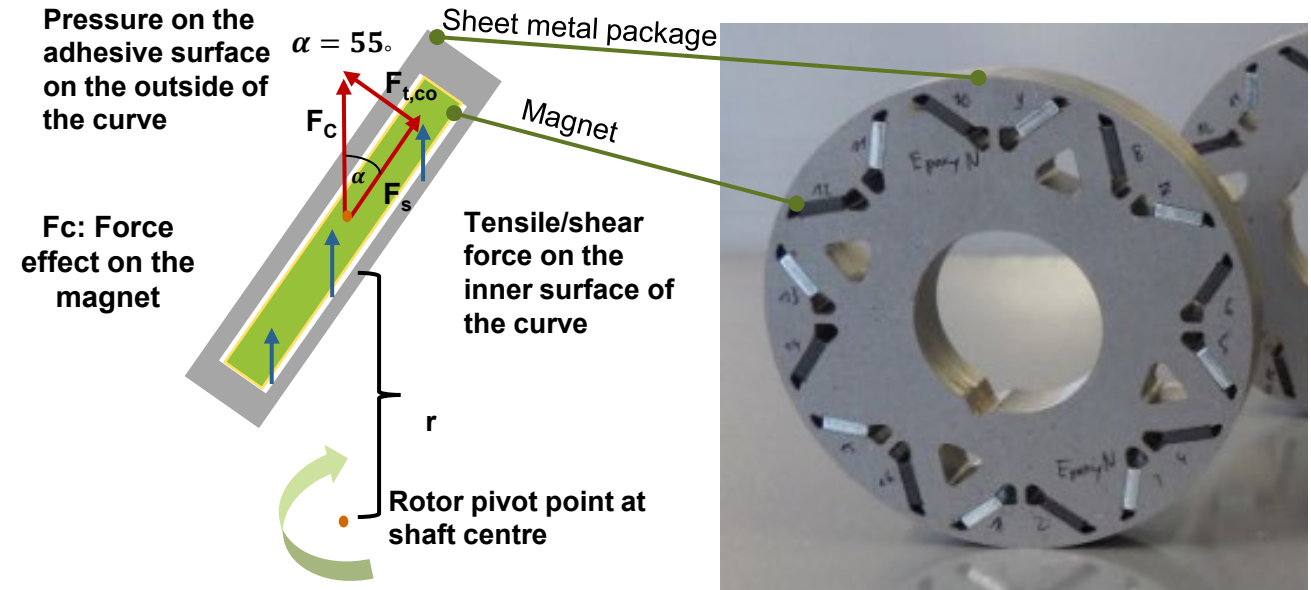
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The shear force F_{shear} and compressive/tensile force $F_{\text{tens,comp}}$ occur, whereby the centrifugal force F_c acts on the magnets at an angle of 55° .

Centrifugal force on shell element

$$F_c = m * r * \omega^2 = \frac{m r 4 \pi^2 n^2}{60^2} = \frac{0,009032 \text{ kg} * 65 \text{ mm} * 4 \pi^2 * 15000^2}{60^2 \text{ s}}$$

$$F_c = 1.448,55 \text{ N}$$

$$\text{With } \omega = \frac{2 \pi n}{60}$$

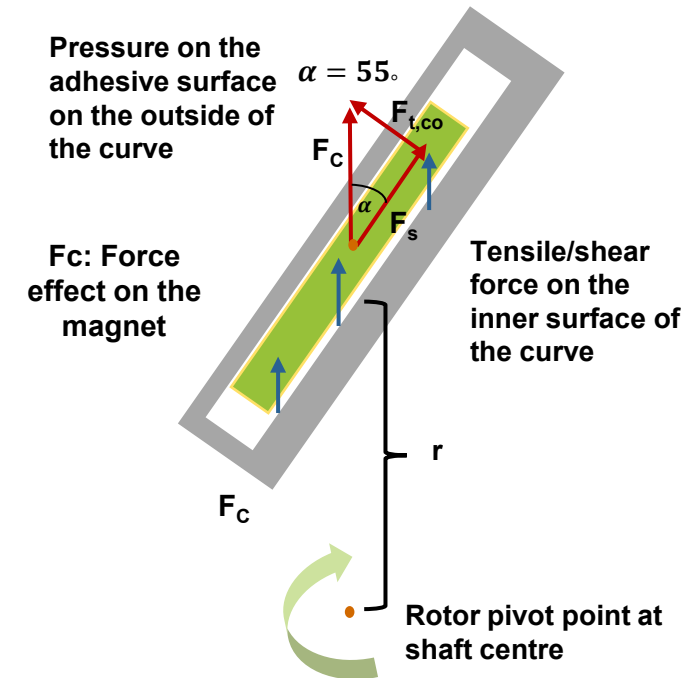
$$m_{\text{magnet}} = \rho_{\text{equiv}} * V_{\text{equiv}} = 14,23 \text{ mm} * 4 \text{ mm} * 21,3 \text{ mm} * 7600 * 10^{-9} \frac{\text{kg}}{\text{mm}^3}$$

$$m_{\text{magnet}} = \rho_{\text{equiv}} * V_{\text{equiv}} = 0,009032 \text{ kg}$$

The force components $F_{t,c}$ and F_s act on the solenoid.

$$F_{\text{tens/comp}} = \sin(55^\circ) * F_c = 1.286,72 \text{ N}$$

$$F_{\text{shear}} = \cos(55^\circ) * F_c = 900,96 \text{ N}$$



Magnet and pocket data:

- Dimensions Magnet: $b = 14,23 \text{ mm}$; $h = 4 \text{ mm}$; $l = 21,3 \text{ mm}$
- Density Magnet $\rho_{\text{equiv}} = 7600 * 10^{-9} \frac{\text{kg}}{\text{mm}^3}$
- Distance pivot point: $r = 65 \text{ mm}$
- Speed $n = 15000 \text{ rpm}$

The adhesive strengths are calculated with an epoxy resin in all three stress directions. All stresses were measured at $T=23^{\circ}\text{C}$ and are higher than the centrifugal forces.

The adhesive's adhesive forces in the tensile and shear directions must be higher than the two calculated forces in operation in order to guarantee the bond.

$$F_{\text{tens bond}} > F_{\text{tens/comp}}$$

$$F_{\text{tens, bond}} > F_{\text{tens/comp}}$$

$$F_{\text{shear, bond}} > F_{\text{shear}}$$

Data on adhesives:

$$\text{Delo Duopox: } \sigma_{\text{comp}} = 30 \frac{\text{N}}{\text{mm}^2}, \sigma_{\text{tens}} = 32 \frac{\text{N}}{\text{mm}^2}, \tau_{\text{shear}} = 19 \frac{\text{N}}{\text{mm}^2}$$

$$\text{ThreeBond TB1530: } \sigma_{\text{comp}} = \text{n.a.}, \sigma_{\text{tens}} = 4,5 \frac{\text{N}}{\text{mm}^2}, \tau_{\text{shear}} = 5,8 \frac{\text{N}}{\text{mm}^2}$$

Delo Duopox

$$F_{\text{comp, bond}} =$$

$$F_{\text{tens, bond}} =$$

$$F_{\text{shear, bond}} =$$

$$S_t =$$

ThreeBond TB1530

$$F_{\text{comp, bond}} =$$

$$F_{\text{tens, bond}} =$$

$$F_{\text{shear, bond}} =$$

$$S_t =$$

The adhesive strengths are calculated with an epoxy resin in all three stress directions. All stresses were measured at T=23°C and are higher than the centrifugal forces.

The adhesive's adhesive forces in the tensile and shear directions must be higher than the two calculated forces in operation in order to guarantee the bond.

$$\begin{aligned}
 F_{\text{tens, bond}} &> F_{\text{tens, comp}} & F_{\text{tens, comp}} &= \sin(55^\circ) * F_c = 1.286,72 \text{ N} \\
 F_{\text{comp, bond}} &> F_{\text{tens, comp}} & F_{\text{shear}} &= \cos(55^\circ) * F_c = 900,96 \text{ N} \\
 F_{\text{shear, bond}} &> F_{\text{shear}} & l_o * b &= 21,3 \text{ mm} * 14,23 \text{ mm}
 \end{aligned}$$

Data on adhesives:

Delo Duopox: $\sigma_{\text{comp}} = 30 \frac{\text{N}}{\text{mm}^2}$, $\sigma_{\text{tens}} = 32 \frac{\text{N}}{\text{mm}^2}$, $\tau_{\text{shear}} = 19 \frac{\text{N}}{\text{mm}^2}$

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Delo Duopox

$$F_{\text{comp, bond}} = \sigma_c * l_o * b = 30 \frac{\text{N}}{\text{mm}^2} * 21,3 \text{ mm} * 14,23 \text{ mm} = 9.093 \text{ N}$$

$$F_{\text{tens, bond}} = \sigma_t * l_o * b = 32 \frac{\text{N}}{\text{mm}^2} * 21,3 \text{ mm} * 14,23 \text{ mm} = 9.699 \text{ N}$$

$$F_{\text{shear, bond}} = \tau_s * l_o * b = 19 \frac{\text{N}}{\text{mm}^2} * 21,3 \text{ mm} * 14,23 \text{ mm} = 5.759 \text{ N}$$

$$S_t = 7 \quad = 9093 \text{ N} / 1.286,72 \text{ N}$$

ThreeBond TB1530

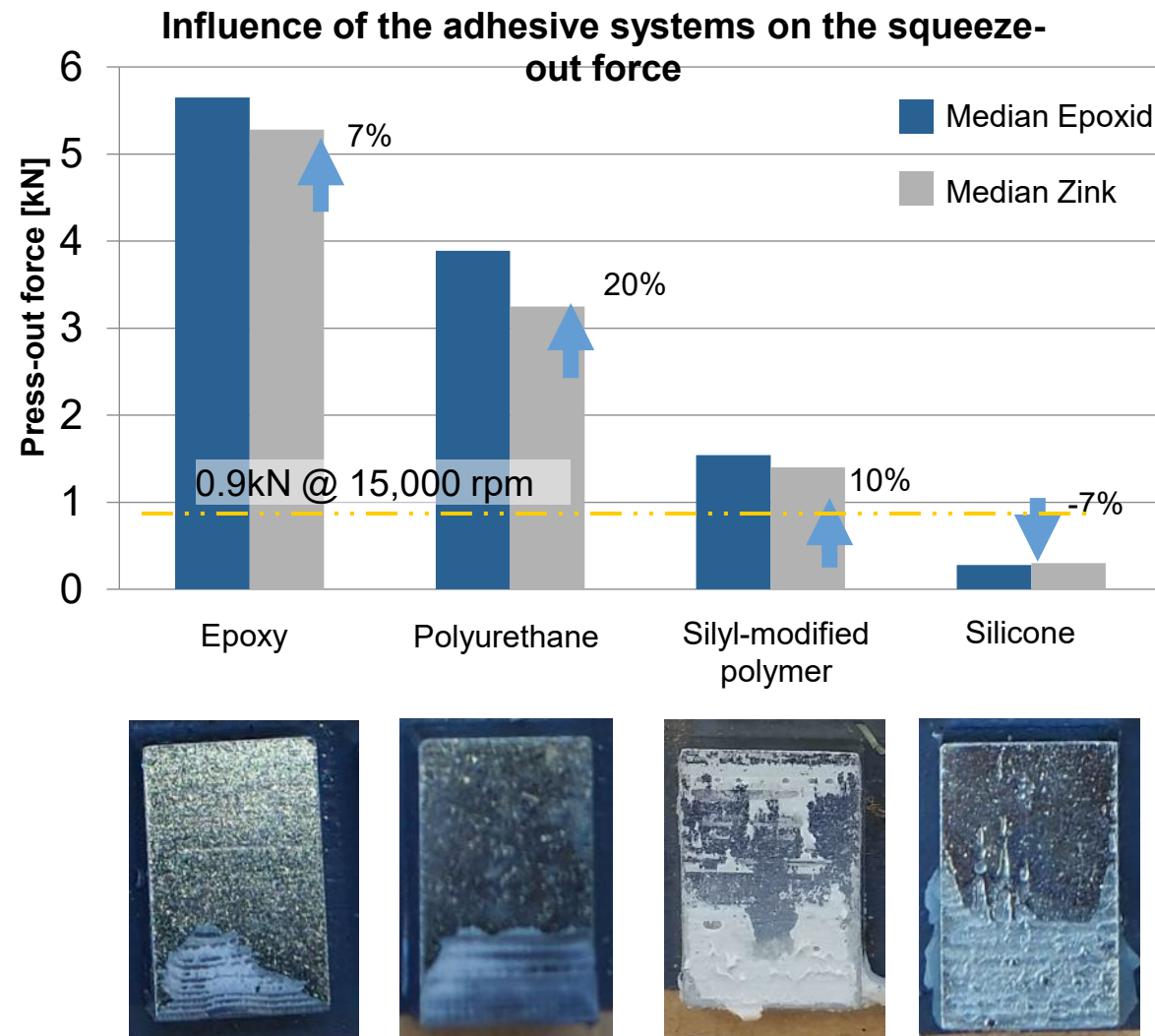
$$F_{\text{comp, bond}} = \sigma_c * l_o * b = \text{n.a.}$$

$$F_{\text{tens, bond}} = \sigma_t * l_o * b = 4,5 \frac{\text{N}}{\text{mm}^2} * 21,3 \text{ mm} * 14,23 \text{ mm} = 1.364 \text{ N}$$

$$F_{\text{shear, bond}} = \tau_s * l_o * b = 5,8 \frac{\text{N}}{\text{mm}^2} * 21,3 \text{ mm} * 14,23 \text{ mm} = 1.758 \text{ N}$$

$$S_t = 1,06 \quad = 1364 \text{ N} / 1.286,72 \text{ N}$$

Both the adhesive systems and the surface coatings have a significant influence on the result.



Surface of the coatings

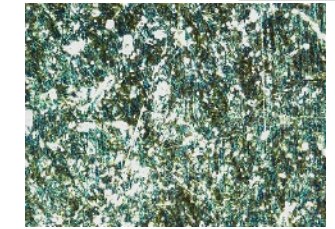
Epoxy



Rz 9.38 μm

Ra 1.71 μm

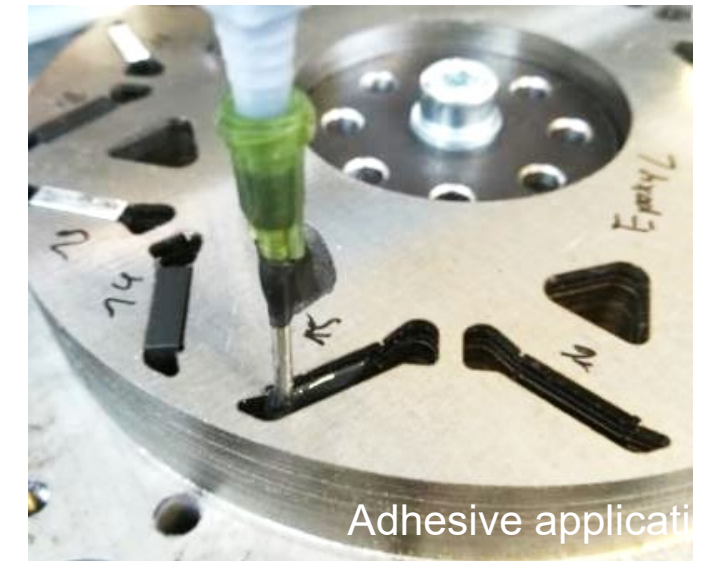
Zinc



Rz 3.48 μm

Ra 0.65 μm

Gluing



Part 2 (stator) of the specialisation builds on the understanding of lecture unit 4.

Relevant topics for preparation in self-study

Rotor assembly

Application example:

- Design of an adhesive joint for an IPM rotor

Methods:

- Strength of materials
- Dynamics

Technical functions:

- Fuses for high-speed concepts
- Qualification of connection technologies

Lecture unit
4



Hard magnetic rotor of a permanent magnet synchronous machine

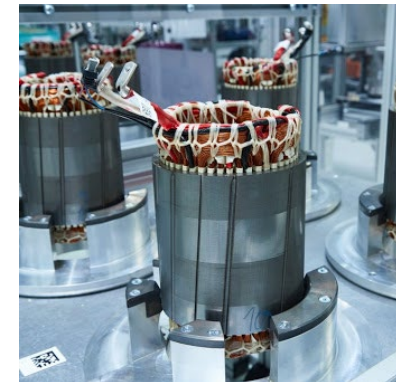
Image: Swiss Federal Office of Energy SFOE (Switzerland)

Stator winding

Relevant topics:

- Filling factors
- Semi-finished wire products
- Winding specification
- Winding scheme
- Winding process
- Wire tension control

Lecture unit
5+6



Stator of a PSM with pull-in winding in automotive engineering

Picture: BMW

Task 1: Tolerance analysis of semi-finished wire products

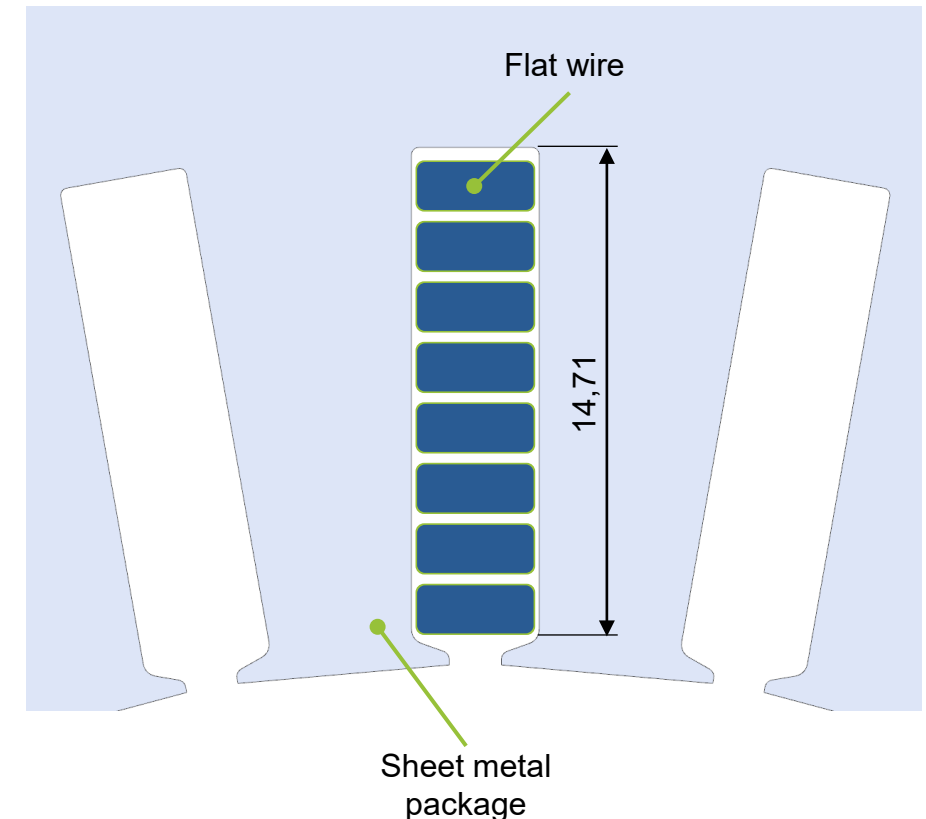
On the right-hand side you can see a section of a stator laminated core fitted with flat wires. The stator slot has a length of 14.71 mm and is 3.52 mm wide. You know from the lecture that the nominal external dimensions change due to different enamelled insulation. Assume that there is a mounting clearance of 0.1 mm between the laminated core and the conductor. From the information given, calculate the minimum installation clearance between two conductors with different degrees of insulation (grade 1 and grade 2).

Given:

- 8 conductors per slot
- Nominal wire width: 3.15 mm
- Nominal wire thickness: 1.60 mm
- Assembly clearance between conductor and laminated core: 0.1 mm

Degree	Increase in dimensions mm		
	Minimum value	Nominal value	Maximum value
1	0,06	0,085	0,11
2	0,12	0,145	0,17

Source: DIN EN 60317-0-2



Solution: Tolerance analysis of semi-finished wire products

On the right-hand side you can see a section of a stator laminated core fitted with flat wires. The stator slot has a length of 14.71 mm and is 3.52 mm wide. You know from the lecture that the nominal external dimensions change due to different enamelled insulation. Assume that there is a mounting clearance of 0.1 mm between the laminated core and the conductor. From the information given, calculate the minimum installation clearance between two conductors with different degrees of insulation (grade 1 and grade 2).

Wanted:

- Installation clearance between two conductors with insulation grade 1
- Installation clearance between two conductors with insulation grade 2

Solution: Tolerance analysis of semi-finished wire products

On the right-hand side you can see a section of a stator laminated core fitted with flat wires. The stator slot has a length of 14.71 mm and is 3.52 mm wide. You know from the lecture that the nominal external dimensions change due to different enamelled insulation. Assume that there is a mounting clearance of 0.1 mm between the laminated core and the conductor. From the information given, calculate the minimum installation clearance between two conductors with different degrees of insulation (grade 1 and grade 2).

Wanted:

- Installation clearance between two conductors with insulation grade 1
- Installation clearance between two conductors with insulation grade 2

Grade 1 solution:

- Maximum value of the outer dimensions for grade 1 (conductor thickness): $1.6 \text{ mm} + 0.11 \text{ mm} = 1.71 \text{ mm}$
- Assembly clearance between ladders at grade 1: (groove height - (8 x maximum external dimensions) - 2 x assembly clearance (ladder - sheet metal))
- Mounting clearance between conductors at grade 1: $14.71 - (8 \times 1.71 \text{ mm}) - 2 \times 0.1 \text{ mm} = 0.83 \text{ mm} \rightarrow 0.83 \text{ mm} \div 7 = 0.12 \text{ mm}$

Grade 2 solution:

- Maximum value of the outer dimensions for grade 2 (conductor thickness): $1.6 \text{ mm} + 0.17 \text{ mm} = 1.77 \text{ mm}$
- Assembly clearance between ladders at grade 2: (groove height - (8 x maximum external dimensions) - 2 x assembly clearance (ladder - sheet metal))
- Mounting clearance between conductors at grade 2: $14.71 - (8 \times 1.77 \text{ mm}) - 2 \times 0.1 \text{ mm} = 0.35 \text{ mm} \rightarrow 0.35 \text{ mm} \div 7 = 0.05 \text{ mm}$

Task 2: Calculation of copper filling factors

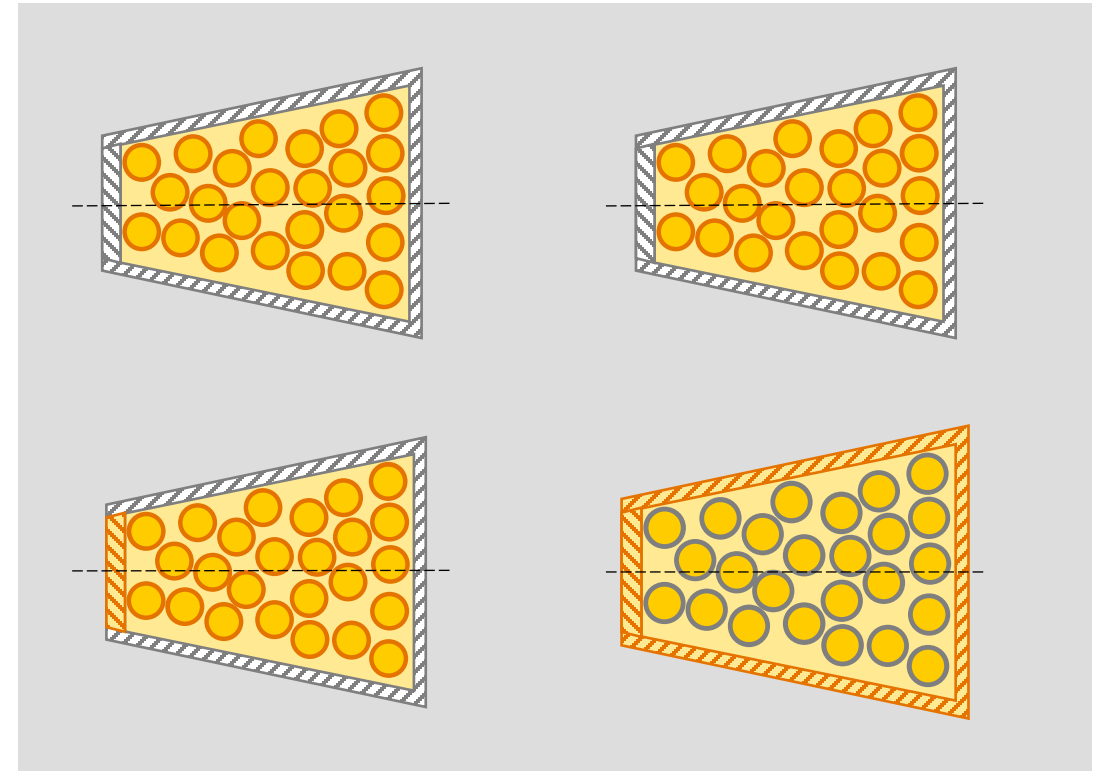
To design an electrical machine, you are to calculate various filling factors. For this purpose, there are four stranded conductors in a slot ($A_{\text{slot}} = 30 \text{ mm}^2$), a slot base insulation is not required. A stranded conductor consists of 20 individual wires with a nominal diameter of 0.5 mm. Due to the applied enamelled insulation, the outer diameter of an insulated wire is 0.54 mm. State the calculated electrical filling factor in per cent! Name three further filling factors and calculate these as well. Assume that there is no slot insulation.

Wanted

■ Calculated electrical filling factor: %

■ Other filling factors

- ...
- ...
- ...



Solution II: Calculation of copper filling factors

To design an electrical machine, you are to calculate various filling factors. For this purpose, there are four stranded conductors in a slot ($A_{\text{Slot}} = 30 \text{ mm}^2$), a slot base insulation is not required. A stranded conductor consists of 20 individual wires with a nominal diameter of 0.5 mm. Due to the applied enamelled insulation, the outer diameter of an insulated wire is 0.54 mm. State the calculated electrical filling factor in per cent! Name three further filling factors and calculate these as well. Assume that there is no slot insulation.

Given

- $A_{\text{Slot}} = 30 \text{ mm}^2$
- $d = 0.5 \text{ mm}$
- $d_i = d_{i \max} = 0.54 \text{ mm}$
- $z_n = 4 \times 20 = 80$

Wanted

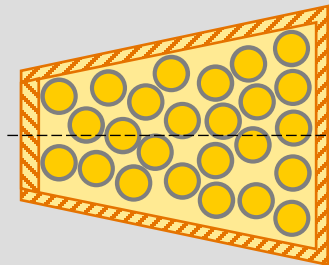
- Electrical filling factor
- Winding filling factor
- Mechanical filling factor
- Slot filling factor

Solution

Electrical filling factor (copper filling factor)

$$k_{\text{elektr}} = \frac{\frac{1}{4} \cdot d^2 \cdot \pi \cdot z_n}{A_n} = \frac{\frac{1}{4} \cdot (0.5 \text{ mm})^2 \cdot \pi \cdot 80}{30 \text{ mm}^2} = 0.5236 = 52.4\%$$

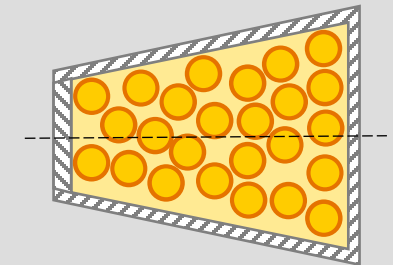
- z_n Number of wires per slot
- d Nominal diameter of the bare wire
- A_n Total slot cross section



Mechanical filling factor

$$k_{\text{mech}} = \frac{\frac{1}{4} \cdot d_i^2 \cdot \pi \cdot z_n}{A_{n i}} = \frac{\frac{1}{4} \cdot (0.54 \text{ mm})^2 \cdot \pi \cdot 80}{30 \text{ mm}^2} = 0.6107 = 61.1\%$$

- z_n Number of wires per slot
- d_i Outer diameter of the insulated wire
- $A_{n i}$ Cross Section of the insulated slot



Solution II: Calculation of copper filling factors

To design an electrical machine, you are to calculate various filling factors. For this purpose, there are four stranded conductors in a slot ($A_{\text{Slot}} = 30 \text{ mm}^2$), a slot base insulation is not required. A stranded conductor consists of 20 individual wires with a nominal diameter of 0.5 mm. Due to the applied enamelled insulation, the outer diameter of an insulated wire is 0.54 mm. State the calculated electrical filling factor in per cent! Name three further filling factors and calculate these as well. Assume that there is no slot insulation.

Given

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- $d = 0.5 \text{ mm}$
- $d_i = d_{i \max} = 0.54 \text{ mm}$
- $z_n = 4 \times 20 = 80$

Wanted

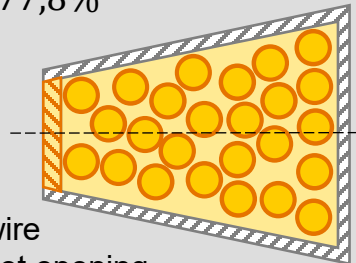
- Electrical filling factor
- Winding filling factor
- Mechanical filling factor
- Slot filling factor

Solution

Winding filling factor

$$k_W = \frac{z_n \cdot d_{i \max}^2}{A_{n \text{ eff}}} = \frac{80 \cdot (0.54 \text{ mm})^2}{30 \text{ mm}^2} = 0.778 = 77.8\%$$

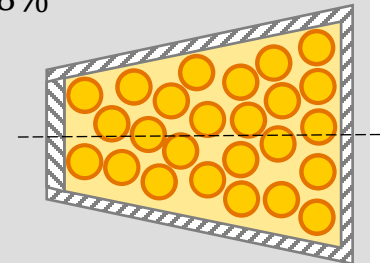
- z_n Number of wires per slot
- $d_{i \max}$ Largest outer diameter of the insulated wire
- $A_{n \text{ eff}}$ Cross section of the insulated slot and slot opening



Slotfilling factor

$$k_N = \frac{z_n \cdot d_i^2}{A_{n i}} = \frac{80 \cdot (0.54 \text{ mm})^2}{30 \text{ mm}^2} = 0.778 = 77.8\%$$

- z_n Number of wires per slot
- d_i Outer diameter of the insulated wire
- $A_{n i}$ Cross Section of the insulated slot



Fill factor of an electrical winding may be calculated using different methods and attention must be paid to the definition of the fill factor in use.

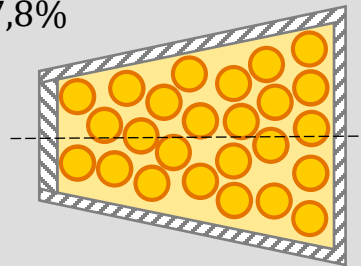
With respect to Wire Diameter

Slot fill factor

- Insulated wire diameters to insulated closed slot area ratio
- Considers Slot liners (Insulation), Wedges, Deck Slider
- Application: e.g. Mechanical design and feasibility studies

$$k_N = \frac{z_n \cdot d_i^2}{A_{ni}} = \frac{80 \cdot (0,54\text{mm})^2}{30\text{mm}^2} = 0,778 = 77,8\%$$

z_n Number of wires per slot
 d_i Outer diameter of the insulated wire
 A_{ni} Cross Section of the insulated slot

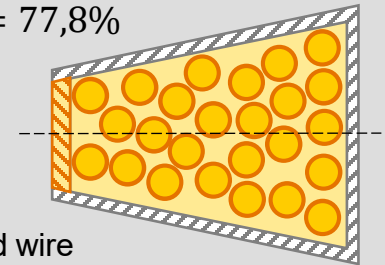


Winding filling factor

- Insulated wire diameters to insulated open slot area ratio
- Considers i.g. Slot liners (Insulation) or Wedges
- Application: e.g. Process design

$$k_W = \frac{z_n \cdot d_{i\max}^2}{A_{n\text{eff}}} = \frac{80 \cdot (0,54\text{mm})^2}{30\text{mm}^2} = 0,778 = 77,8\%$$

z_n Number of wires per slot
 $d_{i\max}$ Largest outer diameter of the insulated wire
 $A_{n\text{eff}}$ Cross section of the insulated slot and slot opening



Fill factor of an electrical winding may be calculated using different methods and attention must be paid to the definition of the fill factor in use.

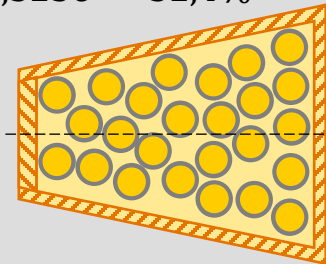
With respect to cross-sectional Wire Area (more relevant)

Electrical filling factor (copper filling factor)

- Copper diameters to full slot area ratio
- Considers no insulation
- Application: e.g. Efficiency Optimization

$$k_{elektr} = \frac{\frac{1}{4} \cdot d^2 \cdot \pi \cdot z_n}{A_n} = \frac{\frac{1}{4} \cdot (0,5mm)^2 \cdot \pi \cdot 80}{30mm^2} = 0,5236 = 52,4\%$$

z_n Number of wires per slot
 d Nominal diameter of the bare wire
 A_n Total slot cross section

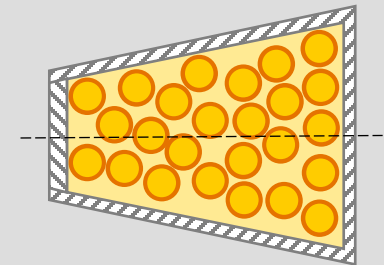


Mechanical filling factor

- Insulated wire diameters to insulated slot area ratio
- Considers Slot liners (Insulation), Wedges, Deck Sliders
- Application: e.g. Thermal Calculations

$$k_{mech} = \frac{\frac{1}{4} \cdot d_i^2 \cdot \pi \cdot z_n}{A_{ni}} = \frac{\frac{1}{4} \cdot (0,54mm)^2 \cdot \pi \cdot 80}{30mm^2} = 0,6107 = 61,1\%$$

z_n Number of wires per slot
 d_i Outer diameter of the insulated wire
 A_{ni} Cross Section of the insulated slot



The contents of the lecture unit can be further deepened with the following the following specialised literature:

Essays

- Bickel, B.; et al. (2014): "Analysis of the optimisation potential for increasing the copper fill factor in electrical machines". Journal article
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